

Literature-Implied Partial Boundary for arXiv:1304.7575

FA/Banach run fa.banach_001

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Status

This is a `literature_implied_answer` (partial characterization / negative boundary example) packet. It does not claim a new proof, and it does not claim a complete classification of all Banach spaces satisfying the stochastic convolution hypothesis (H_p) in van Neerven–Veraar–Weis [1]. It records that the later paper van Neerven–Veraar–Weis [2] gives a substantial positive criterion, an equivalence reducing the Banach-lattice case to deterministic ℓ^1 -boundedness, and a counterexample showing that UMD type 2 is not sufficient.

Source Problem

In the stochastic maximal L^p -regularity section of [1], the authors formulate Hypothesis (H_p) as an R -boundedness condition for a family of stochastic convolution operators. They prove that if X is isomorphic to a closed subspace of an $L^q(\mathcal{O})$ space with $q \in [2, \infty)$, then (H_p) holds for all $p \in (2, \infty)$; the same result holds for $p = q = 2$. Immediately after this theorem they write that it is open to describe the class of Banach spaces X to which the theorem can be extended. They also record a sufficient Banach-function-space condition from earlier work.

Later Literature

The later paper [2] studies the R -boundedness of the same stochastic convolution families. Its Theorem 7.1 states that, for Banach spaces of type 2 and $p \in [2, \infty)$, several R -boundedness formulations for stochastic convolution families are equivalent. In the Banach lattice case these are also equivalent to ℓ^1 -boundedness of an associated deterministic convolution family on $L^{p/2}(\mathbb{R}_+; X^2)$, where X^2 is the 2-concavification of X .

Theorem 7.2 of [2] gives the positive criterion: if X is a Banach lattice with type 2 and the dual of X^2 is an HL lattice, then for every Hilbert space H and every $p \in (2, \infty)$ the family

$$\{S_k^H : k^2 \in \tilde{\mathcal{K}}\}$$

is R -bounded from

$$L_{\mathcal{F}}^p(\mathbb{R}_+ \times \Omega; \gamma(H, X)) \quad \text{to} \quad L^p(\mathbb{R}_+ \times \Omega; X).$$

The following corollary applies this result to the kernel class used in the stochastic maximal regularity argument.

Negative Boundary

Section 8 of [2] explicitly says that it had been open whether this R -boundedness holds for all UMD Banach spaces with type 2, and in particular for all UMD Banach lattices with type 2. The paper answers this negatively.

Theorem 1 (van Neerven–Veraar–Weis, Theorem 8.2). *For every $1 < p < \infty$, the deterministic family $\{T_k : k \in \tilde{\mathcal{K}}\}$ fails to be ℓ^1 -bounded on $L^p(\mathbb{R}_+; \ell^1(\ell^2))$. Consequently, for every $2 < p < \infty$ the stochastic convolution family $\{S_k^H : k^2 \in \tilde{\mathcal{K}}\}$ fails to be R -bounded from*

$$L^p_{\mathcal{F}}(\mathbb{R}_+ \times \Omega; \gamma(H, \ell^2(\ell^4))) \quad \text{to} \quad L^p(\mathbb{R}_+ \times \Omega; \ell^2(\ell^4)).$$

Thus the broad source question cannot be answered by the condition “UMD and type 2” alone. Even within UMD Banach lattices of type 2, the space $\ell^2(\ell^4)$ is outside the stochastic convolution class considered in the theorem.

Conclusion

For the source problem asking for the class of Banach spaces to which the L^q stochastic maximal regularity theorem extends, [2] provides the following literature-implied partial answer:

1. in Banach lattices the question is equivalent to a deterministic ℓ^1 -boundedness problem on the 2-concavification;
2. Banach lattices of type 2 whose 2-concavification has dual an HL lattice give a positive class;
3. UMD type 2 is not sufficient, because $X = \ell^2(\ell^4)$ is a counterexample.

The full classification requested in [1] remains broader than this packet.

Search Evidence

The run indexes were searched for 1304.7575, Hypothesis (H_p), stochastic maximal regularity, R -bounded stochastic convolution, ℓ^1 -boundedness, UMD type 2, and $\ell^2(\ell^4)$. Existing packets covered adjacent stochastic Weiss and UMD-lattice themes but not this source problem. A web/arXiv check on June 15, 2026 identified [2] as the direct follow-up source.

Human Review Recommendation

Review as a provenance/status packet. The key points to check are whether a later paper gives a full intrinsic classification of all Banach spaces satisfying (H_p) , and whether theorem numbering differs in the published versions of the two arXiv papers.

References

- [1] J. van Neerven, M. Veraar, and L. Weis, *Stochastic integration in Banach spaces – a survey*, arXiv:1304.7575.
- [2] J. van Neerven, M. Veraar, and L. Weis, *On the R -boundedness of stochastic convolution operators*, arXiv:1404.3353.